

Q.Explain the Eigenvalue and eigenvector in detail along with some examples. Its role in PCA.
Ans.

Scenario 1

Suppose we have some pebbles and we throw those pebbles, pebbles are scattered in different direction here eigen vectors tells us in which mainly pebbles are scattered and eigen values tells us how long in terms of magnitude these pebbles are stretched.

Scenario 2

Suppose we draw height and weight of sample containing 10000 people. Person with more height tends to have more weight. Now if we will plot this eigen vector will tell us direction of relationship between height and weight. Eigen value will tell us how much variation exist in that direction.

Scenario 3

Considering we have thousands of images of Humans. Here eigen vector will tell us important features/patterns in the images. While eigen value will tell us how much variance is explained in these patterns.

In PCA eigen vectors determine the direction of principal components while eigen value represents the amount of variance explained by each principal components. Eigen vector corresponding to largest eigen value becomes PC1. Eigen vector corresponding to second largest eigen value become PC2 and so on.